Ruşen Birben

AI Researcher & Engineer

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AI and Data Engineering graduate from Istanbul Technical University and independent researcher with METU's Applied NLP Group. Research interests center on LLM reliability and interpretability, with experience in Turkish NLP, machine unlearning, and LLM applications for finance.

EDUCATION

B.Sc. AI and Data Engineering 2022 - 2025
Istanbul Technical University GPA: 3.61 / 4.00
Graduated at 09.07.2025

Electronics and Communication Engineering 2020 - 2022
Istanbul Technical University
Transferred to AI & Data Engineering

SCHOLARSHIPS & AWARDS

1416 YLSY Scholarship · M.Sc. & Ph.D. 2026
Fully funded government scholarship for M.Sc. and Ph.D. study in Artificial Intelligence and Big Data in the USA, awarded through a national selection exam.

1st Place · Agenthon 2026, Track 3 (Simulation) 2026
Ranked 1st of 55 teams in the simulation track with an event-for-event C++ rebuild of the ABIDES market simulator (see Publications). The win earned an additional team poster presentation at the Agenthon workshop at NeurIPS 2026.

Kaggle Bronze Medal · The Pokémon Company PTCG AI Battle Challenge 2026
Ranked 677th of 6,807 teams (top 10%) in Kaggle's simulation competition, developing and evaluating partially observable game-playing agents with replay-based behavioral cloning, belief-aware Monte Carlo planning, and self-play.

RESEARCH EXPERIENCE

Independent Researcher 03/2026 - Present
METU Applied NLP Group Ankara, TR

- Co-organize the SIGTURK 2027 Shared Task at the SIGTURK 2027 Workshop (ACL 2027) with Çağrı Toraman, Evren Ayberk Munis, and İlker Kesen. Contribute to data creation and evaluation methodology; built the shared-task website and evaluation pipeline.
- Contribute to GemmaTurk, a Turkish LLM, across training-data collection, synthetic instruction and reasoning data generation, mid-training, continual pre-training, and pre-training evaluation.
- Lead METU NLP Connect, producing expert interviews.

AIOps Research Intern 08/2024 - 09/2024
Havelsan Istanbul, TR

- Built LSTM and random-forest prototypes for log-based anomaly detection on HDFS logs for predictive maintenance; the LSTM reached 0.96 F1 and 99% accuracy on the test set.

NLP Research Intern 07/2024 - 08/2024
ITU NLP Lab Istanbul, TR

- Ran experiments on financial LLMs across sentiment analysis, tool use, reasoning, forecasting, and RAG, feeding into the Kaira capstone.

PUBLICATIONS & RESEARCH CONTRIBUTIONS

Exact First, Then Fast: An Event-for-Event C++ Rebuild of the Agenthon ABIDES Market 2026

Agenthon Workshop @ NeurIPS 2026 - Accepted for in-person poster presentation Competition

- Göktürk Batın Dervişoğlu, Muhammed Faruk Aytin, Muhammed Rüşen Birben.
- Rebuilt the Agenthon ABIDES market simulator in C++, preserving event ordering, random-number sequences, and reference outputs. Validated exact agreement across 6,900 randomized comparisons and achieved a 1,299× simulation-loop speedup on the largest public benchmark.
- Atlanta, USA, December 12, 2026.

Turkish Cultural Knowledge Dataset 2026

MRL Workshop @ EMNLP 2026 - Accepted dataset contribution; joint benchmark paper in preparation

Shared Task

- Led the source-grounded agent pipeline and data preparation for a Turkish cultural knowledge dataset spanning 15 domains.
- Screened 1,200 candidate question-answer pairs; selected 500 for sociological review by Büşra Tufan and prepared the final 300-pair submission.
- Reviewed English translations; supervised by Çağrı Toraman.

SELECTED PROJECTS

Kaira 2025

Undergraduate Capstone: Financial Reasoning LLM

- Fine-tuned Qwen-based LLMs with GRPO, Unsloth, and LoRA, reaching 58.08% FinQA accuracy with Qwen3-4B, up from 47.48% with Qwen2.5-3B; ran FinQA and GSM8K experiments on reward design and financial reasoning. Developed an Ollama-based chat application.

Tobor 2024 - 2025

Autonomous Warehouse Robot Code Video

- Implemented the box pickup/drop-off mechanism for a team-built ROS2 warehouse robot with navigation and YOLO perception; contributed to data annotation.

AIZheimer 2024

Machine Unlearning in Stable Diffusion Code

- Adapted Forget-Me-Not unlearning to Stable Diffusion 2.1, implementing attention controllers and testing loss objectives to erase target concepts while preserving unrelated generation.

Biting The Bytes 2024

Turkish Diacritic Restoration Code Demo

- Led ByT5-small fine-tuning with PEFT/LoRA for byte-level Turkish diacritic restoration, reaching 96.18% word-level accuracy on the test set and tying for 2nd of 17 teams in the course's Kaggle competition; co-deployed the model on Hugging Face Spaces.

To-AI-or-Not-to-AI 2023

GPT Detector Code Demo

- Built an ensemble of three fine-tuned RoBERTa detectors combined by majority vote, reaching 94.21% accuracy on the held-out test split of a 26,270-example dataset prepared with teammates.

PROFESSIONAL EXPERIENCE

Co-Founder & Technical Lead
08/2025 - 04/2026

Fiction Studios Ankara, TR

- Co-founded Fiction Studios at Bilkent Cyberpark; led technical development and coordinated three additional contributors. Built a multilingual LLM-powered museum and tourism guide and pitched it to customers and investors.

Junior AI Engineer
03/2025 - 07/2025

CyberQuote Istanbul, TR

- Built RAG pipelines that retrieve up-to-date context from fragmented internal sources and PDF financial reports, and agent pipelines for internal financial LLM tools. Also developed news-sentiment pipelines and web integrations at a Phillip Capital subsidiary.

TECHNICAL SKILLS

Languages Python / JavaScript/TypeScript / SQL / C/C++

Frameworks & Tools PyTorch / TensorFlow / Transformers / PEFT / LoRA / Unsloth / Scikit-learn / Pandas / NumPy / ROS2 / YOLO

Specialties NLP / LLMs / RAG / Machine Unlearning / Model Evaluation

COURSES / LANGUAGES

Deep Learning Specialization DeepLearning.AI · Coursera

AWS Cloud Technical Essentials AWS · Coursera

Turkish Native **English** TOEFL iBT 109/120 (November 8, 2025)

Interests: Coffee / Books / Photography / Anime & Manga